

ACTIVITY 05

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1. Geospatial Data Visualization

Dataset

City	Latitude	Longitude	Population (Millions)
Chennai	13.08	80.27	11
Bangalore	12.97	77.59	13
Hyderabad	17.38	78.48	10
Mumbai	19.07	72.87	20
Delhi	28.61	77.21	32
Pune	18.52	73.85	7
Kolkata	22.57	88.36	15
Ahmedabad	23.02	72.57	8
Jaipur	26.91	75.79	4
Kochi	9.93	76.26	2

Questions

CODE:

```
# Install package if required #  
install.packages("ggplot2")  
library(ggplot2)  
# Dataset  
city_data <- data.frame(  
  City=c("Chennai","Bangalore","Hyderabad","Mumbai","Delhi",  
    "Pune","Kolkata","Ahmedabad","Jaipur","Kochi"),
```

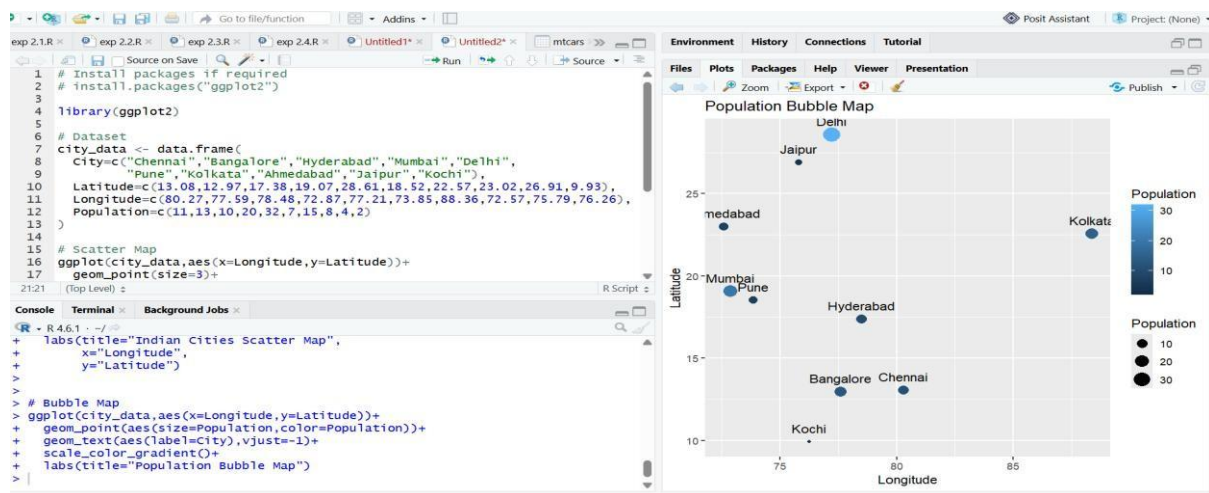
```

Latitude=c(13.08,12.97,17.38,19.07,28.61,
           18.52,22.57,23.02,26.91,9.93),
Longitude=c(80.27,77.59,78.48,72.87,77.21,
           73.85,88.36,72.57,75.79,76.26),
Population=c(11,13,10,20,32,7,15,8,4,2)
)
# Scatter + Bubble Map ggplot(city_data,
aes(x=Longitude,y=Latitude))+
geom_point(aes(size=Population,
               color=Population))+
geom_text(aes(label=City),
          vjust=-1)+ labs(title="Indian
Cities Population Map",
x="Longitude",    y="Latitude")

```

Q1. Create a scatter map showing all cities using longitude and latitude.

Q2. Plot a bubble map where bubble size represents population.



2. Network Visualization

Dataset

From	To
A	B

A	C
B	D
C	D
C	E
D	E
E	F
F	G
G	H
H	A

Questions

CODE:

```
# Install package if required
# install.packages("igraph")
library(igraph) # Dataset
edges <- data.frame(
  From=c("A","A","B","C","C","D","E","F","G","H"),
  To=c("B","C","D","D","E","E","F","G","H","A")
)
# Create Network network <-
graph_from_data_frame(edges)
# Network Graph
plot(network,
  layout=layout_in_circle,
  vertex.color="lightblue",
  vertex.size=30,
```

```
vertex.label.color="black",
```

```
main="Network Graph") #
```

Degree of nodes

```
degree(network)
```

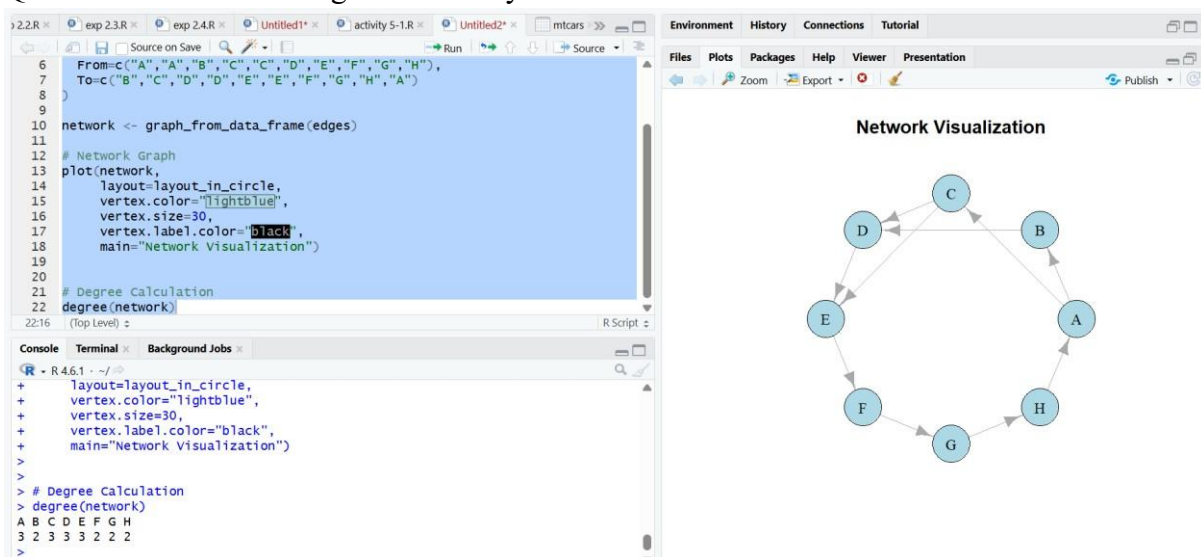
Q1. Create a network graph using the igraph package.

Q2. Display node labels and customize node colors.

Q3. Calculate the degree of each node

Q4. Identify the node with the highest number of connections.

Q5. Plot the network using a circular layout.



3. Temporal Data Representation

Dataset

Month	Sales (₹ Lakhs)
Jan	120
Feb	135
Mar	150
Apr	170
Month	Sales (₹ Lakhs)

May	190
Jun	220
Jul	210
Aug	230
Sep	250
Oct	270
Nov	290
Dec	320

Questions

CODE

```
# Dataset month <-
c("Jan","Feb","Mar","Apr",
  "May","Jun","Jul","Aug",
  "Sep","Oct","Nov","Dec") sales <-
c(120,135,150,170,190,220,
  210,230,250,270,290,320)

# Line Chart
plot(sales,
  type="o",
  pch=19,
  xaxt="n",
  xlab="Month",
  ylab="Sales
(Lakhs)",
  main="Monthly
Sales") # Add
```

month labels axis(1,

at=1:12,

labels=month) #

Grid lines grid()

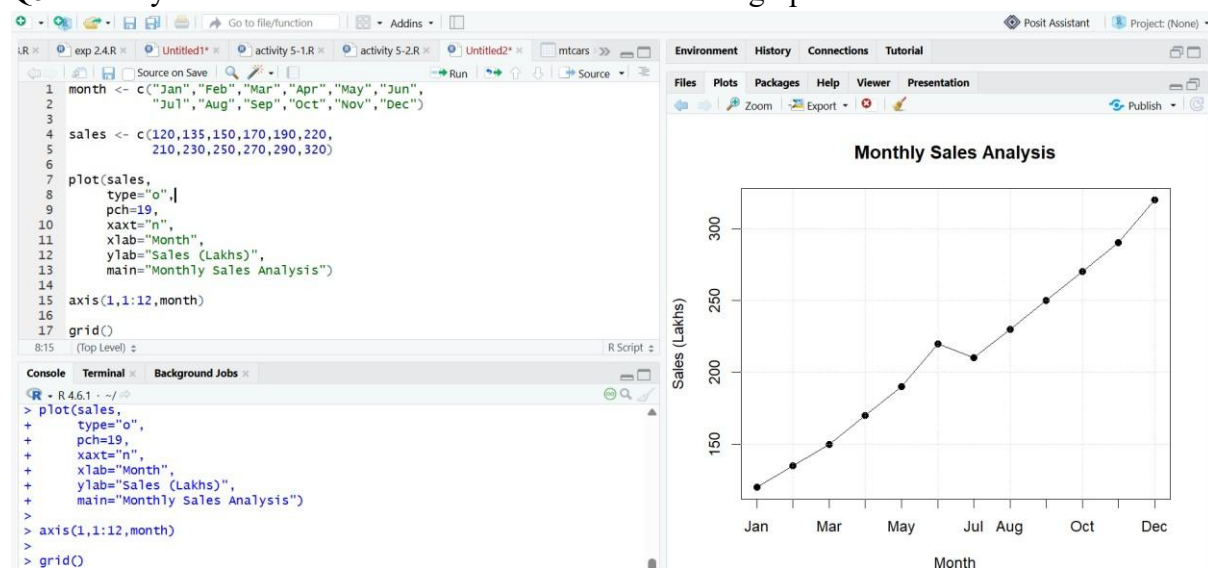
Q1. Create a line chart showing monthly sales.

Q2. Add custom month names using the axis() function.

Q3. Highlight data points using pch=19.

Q4. Display grid lines and proper axis labels.

Q5. Identify the month with the maximum sales from the graph.



4. Multivariate Data Visualization

Dataset

Student	Marks	Attendance (%)	Study Hours	Projects
A	60	75	2	1
B	68	80	3	2
C	74	84	4	2
D	79	88	5	3
E	82	90	5	3

F	85	92	6	4
G	88	94	6	4
H	91	95	7	5
I	94	97	8	5
J	97	99	9	6

Questions

CODE:

```
# Dataset student <-
data.frame(
  Student=c("A","B","C","D","E",
            "F","G","H","I","J"),
  Marks=c(60,68,74,79,82,
          85,88,91,94,97),
  Attendance=c(75,80,84,88,90,
              92,94,95,97,99),
  StudyHours=c(2,3,4,5,5,
              6,6,7,8,9),
  Projects=c(1,2,2,3,3,
            4,4,5,5,6)
)

# Q1 Bubble Plot plot(student$Marks,
student$Attendance,
cex=student$StudyHours/2,   pch=19,
xlab="Marks",   ylab="Attendance
(%)",   main="Student Bubble Plot")
text(student$Marks,
student$Attendance,
```

```

labels=student$Student, pos=3) # Q2
Scatter Plot Matrix pairs(student[,2:5],
main="Student Performance Matrix")
# Q3 Correlation cor(student[,c("Marks",
      "Attendance",
      "StudyHours")])) # Q4

```

Highest Marks Student

```
student[which.max(student$Marks),]
```

Q5 Heatmap

```

heatmap(cor(student[,2:5]),
main="Correlation Heatmap",
scale="none")

```

Q1. Create a bubble plot where:

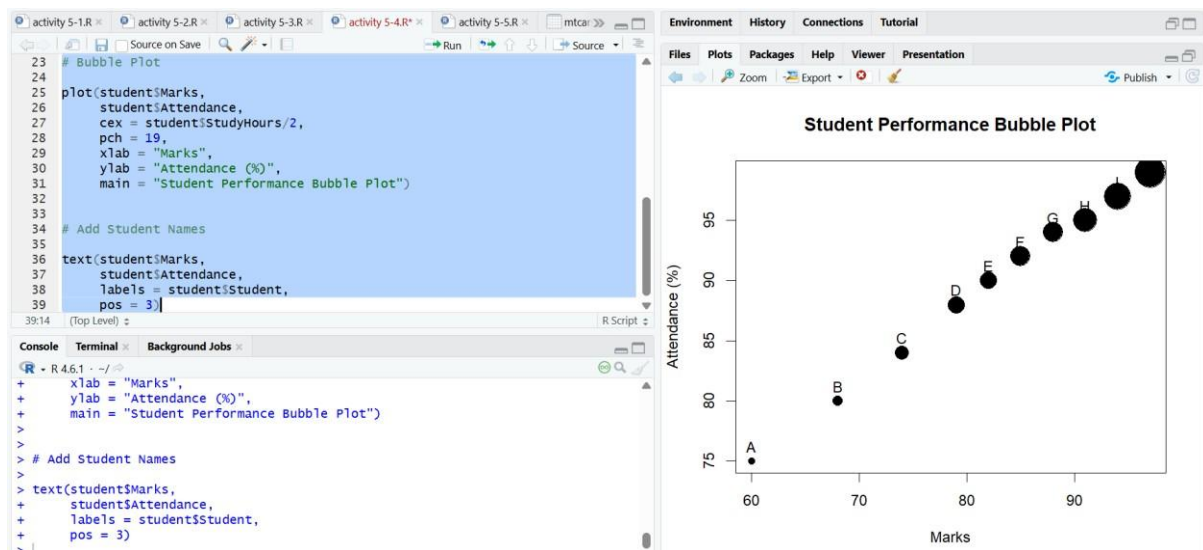
- X-axis = Marks
- Y-axis = Attendance
- Bubble size = Study Hours

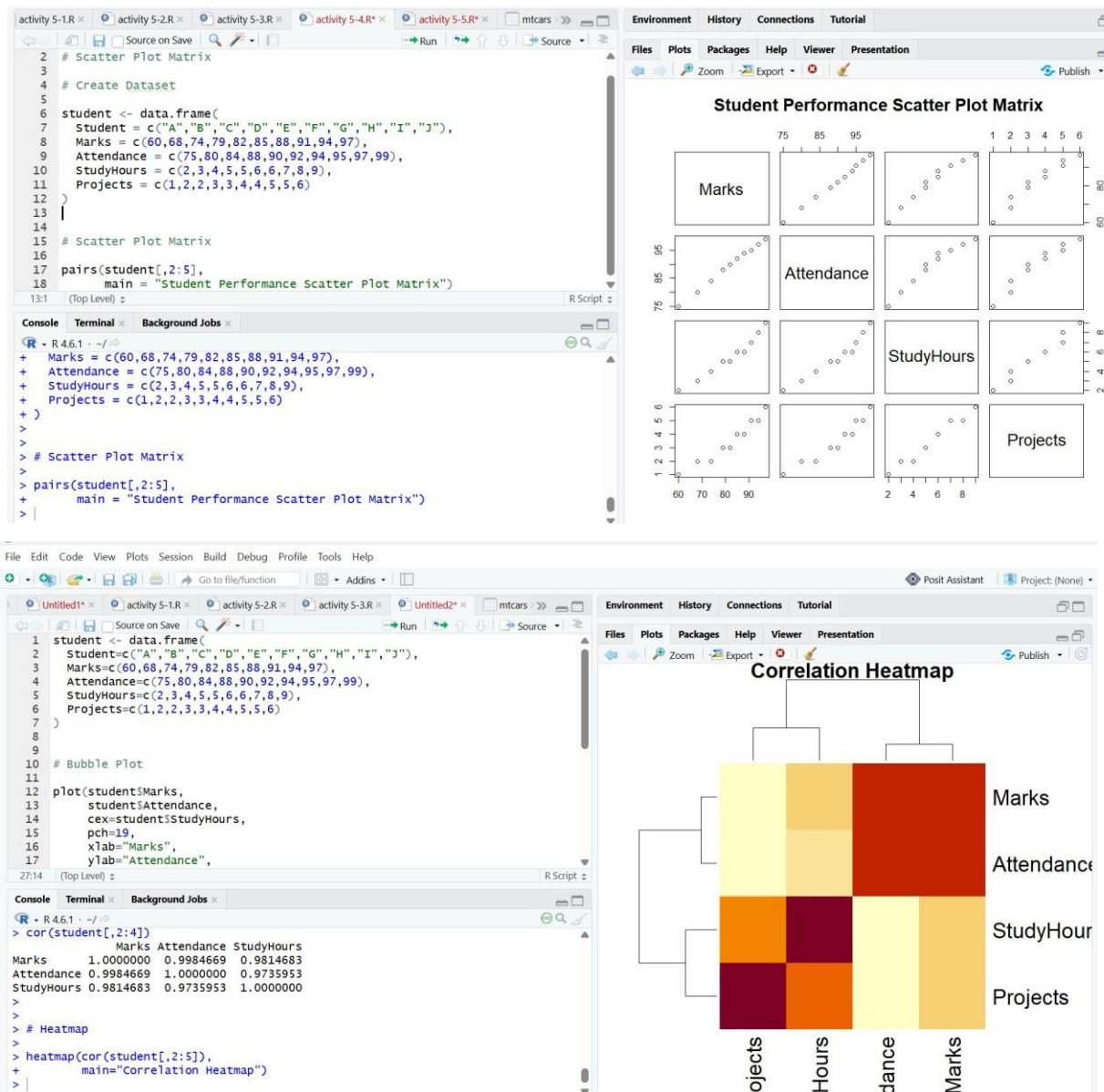
Q2. Produce a scatter plot matrix using the pairs() function.

Q3. Find the correlation among Marks, Attendance, and Study Hours.

Q4. Identify the student with the highest marks and largest bubble.

Q5. Create a heatmap showing the relationships among all numerical variables.





5. Integrated Data Visualization Exercise

Dataset

Year	Sales	Profit	Customers	Branches
2016	150	20	1200	5
2017	170	25	1500	6
2018	200	32	1800	7

2019	240	40	2200	8
2020	260	42	2400	8
2021	300	55	2800	10
2022	340	65	3300	11
2023	390	72	3800	12
2024	430	82	4200	13
2025	480	95	4700	15

Questions

CODE

```
# Dataset company <-
data.frame(
  Year=c(2016,2017,2018,2019,2020,
        2021,2022,2023,2024,2025),
  Sales=c(150,170,200,240,260,
        300,340,390,430,480),
  Profit=c(20,25,32,40,42,
        55,65,72,82,95),
  Customers=c(1200,1500,1800,2200,2400,
        2800,3300,3800,4200,4700),
  Branches=c(5,6,7,8,8,
        10,11,12,13,15)
)

# Q1 Line Chart
plot(company$Year,
  company$Sales, type="o",
  pch=19, xlab="Year",
```

```

ylab="Sales",    main="Yearly
Sales Growth")
# Q2 Bubble Plot plot(company$Sales,
company$Profit,
cex=company$Customers/1000,
pch=19,    xlab="Sales",
ylab="Profit",    main="Sales vs
Profit Bubble Plot")
text(company$Sales,
company$Profit,
labels=company$Year,    pos=3)
# Q3 Correlation Matrix cor(company[,2:5]) #
Q4 Scatter Plot Matrix pairs(company[,2:5],
main="Company Data Scatter Plot Matrix")

```

Q1. Create a temporal line chart showing yearly sales.

Q2. Generate a multivariate bubble plot with:

- X-axis = Sales
- Y-axis = Profit
- Bubble size = Customers

Q3. Compute the correlation matrix for Sales, Profit, Customers, and Branches.

Q4. Create a scatter plot matrix using the pairs() function.

Q5. Interpret the visualizations to determine whether an increase in branches is associated

